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SUPPLEMENTARY MATERIAL The Locked Bimetric Potential and the Constitutive Law

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Abstract
Papers VIII, IX and XII fix the Hassan–Rosen coefficients to $\beta = (1, 0, 16, 0, 1)$ and state that the locked bimetric theory and the SST are two projections of one geometry, so that the bimetric theory's quasi-static limit should give $\mu(x) = x/(x+1)$. We show it does not: Hassan–Rosen's two metrics are not GFD's two chambers. *Negative:* in the standard reduction (decoupling limit, static spherical weak field) the response depends on the enclosed mass only through M/r^3, for every β; no function of M/r^3 equals a function of $x \propto M/r^2$ for all masses and radii; on the branch that matches the linear solution the enhancement is bounded, $g_{\rm tot}/g_N \le 1 + 1/\sqrt{2} < 1.71$, and the other real roots give either $g_{\rm tot}/g_N \le 1/3$ or a mass-independent force growing like r; and the potential vanishes at GFD's parity configuration. Letter 2's Lemma explains why: TetA and TetB are two roots of one metric, so a potential built from $\sqrt{g^{-1}f}$ takes the same value on either root ($g^{-1}f = I$) and cannot see the partition. The obstruction is structural, not specific to the lock. *Diagnostic:* on the frame deformed along a single leg by a ratio w, every potential is $V = A + Bw$, and the deformed leg's share $Bw/(A+Bw)$ obeys exchange closure $s(1/w) = 1 - s(w)$ if and only if $\beta_0 + 2\beta_1 = \beta_4 + 2\beta_3$. It is then exactly $w/(1+w)$. Palindromic coefficients (the $g \leftrightarrow f$ symmetry) satisfy the condition, the locked set gives $V = 49(1+w)$, and $\beta_2 = 16$ cancels. With $w = |a|/a_0$, a the acceleration of the tetrad's time leg, this is $\mu(x)$. The deformation is not a Hassan–Rosen reference metric, the two shares are parts of one deformed frame rather than the two roots (on the roots the potential is 98 and 0), and the energy F is not supplied by the potential. So this reproduces the algebra of the partition, not its two-chamber mechanism. The constitutive law's origin remains the two-root partition under exchange (Letter 2).

Keywords: Hassan–Rosen, bimetric gravity, Vainshtein, Galileon, exchange symmetry, constitutive law, MOND, proof

Setting
The Hassan–Rosen (HR) interaction is $m^2\sqrt{-g}\,V(S)$, $V(S) = \sum_{n=0}^{4}\beta_n e_n(S)$, with $S = \sqrt{g^{-1}f}$ and e_n the elementary symmetric polynomials of its eigenvalues [1]. The lock is $\beta = (1, 0, 16, 0, 1)$ [2, 3]. The SST's scalar sector is $-(a_0^2/8\pi G)F(y)$, $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$, $\mu(x) = F'(x^2) = x/(x+1)$ [6, 7]. The question, listed as open in Papers VIII and IX, is whether the first reduces to the second. Each step is checked by the accompanying script, 27 checks, exact algebra [11], including the decoupling-limit derivation itself, done independently of [10].

Part A: the standard reduction
Theorem 1 (the response depends on density). *In the decoupling limit, with Stückelberg field π and $\Lambda_3^3 = m^2 M_{\rm Pl}$, the h–π mixing has coefficients $d_0 = \beta_0 + 3\beta_1 + 3\beta_2 + \beta_3$, $d_1 = \beta_1 + 2\beta_2 + \beta_3$, $d_2 = \beta_2 + \beta_3$, $d_3 = \beta_3$; for the lock, $(49, 32, 16, 0)$. Around a static mass, with $d_3 = 0$, $\lambda = \pi'/r = -sq/(2Ma_1)$ and $Q = a_2 q/(Ma_1^2)$, the π-equation is*

$$Q^2 s^3 + 6Qs^2 + 6s - 2 = 0, \qquad g_{\rm tot}/g_N = 1 + s, \tag{1}$$

with $q = M_{\rm enc}/4\pi r^3$. The radius enters only through q.

Proof. The identity $\sum_n (-1)^n \beta_n Y_n(1 - \Pi) = \sum_k d_k Y_k(\Pi)$, with $Y_n(X) = \sum_k (-1)^k e_k(X) X^{n-k}$, is checked for general β on an exact random symmetric Π ($Y_4 = 0$ by Cayley–Hamilton). From $-\frac{1}{2}h\mathcal{E}h + h \cdot \sum_k a_k Y_k(\Pi) + h_{00}\rho/2M$ with $h = \mathrm{diag}(A, B, B, B)(r)$ and $\pi(r)$, all three Euler–Lagrange equations are r^{-2} times the derivative of a current (checked), so they integrate exactly. Regularity at the origin gives the π-equation $\frac{3}{2}a_1^2\lambda - (3a_1a_2 + a_3 q/4M)\lambda^2 + (a_2^2 + \frac{2}{3}a_1a_3)\lambda^3 - \frac{1}{6}a_3^2\lambda^5 + a_1 q/4M = 0$, which contains no r, and $g_{\rm tot}/g_N = 1 - 2Ma_1\lambda/q + \frac{2}{3}Ma_3\lambda^3/q$ (checked). Setting $a_3 = 0$ and substituting gives (1). □

$d_1 = 32$ is the Fierz–Pauli combination and $d_3 = 0$ removes the quintic mixing; $d_2 = 16$ keeps the cubic Galileon term (Paper XII states that both vanish).

Theorem 2 (the branches). *Every real root of (1) is $s = 2/D(v)$, $Q = vD(v)/2$, with $D = v^2 + 6v + 6$.*

1. *The branch that matches the linear solution ($v \to 0$: $s = 1/3$, $g_{\rm tot}/g_N = 4/3$, vDVZ) is screened for $Q > 0$ ($s \to 0$). For $Q < 0$ it*

turns at $Q_* = 2 - 2\sqrt{2}$, where $s = 1/\sqrt{2}$, and does not continue past it. On it, $g_{\rm tot}/g_N \le 1 + 1/\sqrt{2} < 1.71$.

2. *Where $D < 0$, $s \le -2/3$, so $g_{\rm tot}/g_N \le 1/3$.*

3. *As $D \to 0^+$ ($v \to -3 \pm \sqrt{3}$), $Q \to 0$ and $s \simeq v/Q \to \infty$. There $g_{\rm tot} \simeq g_N v/Q \propto (M/r^2)(r^3/M) = r$, independent of the mass.*

Proof. Substituting the parametrisation solves (1) identically; the turning points are $v = -2 \pm \sqrt{2}$, $\min D = -3$, and the roots of D are $-3 \pm \sqrt{3}$, where $Q = 0$ (all checked). □

The unbounded roots of (1) do not match the linear solution at large r, and their force grows like r whatever the mass: a self-accelerating-type branch, not the deep-MOND form $\sqrt{GMa_0}/r$. So no branch gives flat rotation curves. Theorem 3 shows this for every branch at once.

Theorem 3 (no change of variable). *No function of M/r^3 equals a function of $x = GM/(a_0 r^2)$ for all M and r, unless both are constant. Since every root of the π-equation is a function of q alone (Theorem 1), the standard response cannot be $1/\mu(x)$ on any branch, for any β.*

Proof. Along $M = Lr^3$ the density is fixed while $x = GLr/a_0$ varies, and $\mathrm{d}[1/\mu(x)]/\mathrm{d}r \neq 0$ (checked). Equivalently, the Vainshtein radius scales as $M^{1/3}$ and the MOND radius $\sqrt{GM/a_0}$ as $M^{1/2}$. □

With $m \sim H_0$, galaxies also lie well inside the Vainshtein radius (∼260 kpc at $10^{10} M_\odot$), where $\mu_{\rm eff} \to 1$ [10]. Dimensionally, the Galileon variable is $\Phi/(mr)^2$ and the MOND variable $\Phi/(mr)$: both small quantities Φ/c^2 and rH_0/c are $\sim 10^{-6}$ at galactic radii, which is why $a_0 \sim cH_0$ is natural, but the potential's strongest interactions select the wrong power.

Theorem 4 (the parity configuration and the polynomial). *At $S = P = \mathrm{diag}(1, -1, -1, -1)$, $e_n(P) = (1, -2, 0, 2, -1)$ and $V(P) = \beta_0 - \beta_4 = 0$. On $S = \mathrm{diag}(1, s, s, s)$ the locked potential is $1 + 48s + 48s^2 + s^3$ (98, 1, 0 at $s = +1, 0, -1$), not the coupling polynomial $1 + 4s + 16s^2$ (21, 17, 13).*

Proof. Direct evaluation (checked). □

So GFD's two roots (one metric, $S = P$) and HR's two metrics are different objects, and the potential's values are not the polynomial's.

Theorem 5 (Hassan–Rosen is root-blind). *TetA $= e$ and TetB $= Pe$ give the same metric, $g = e^{\mathsf{T}} \eta e = (Pe)^{\mathsf{T}} \eta (Pe)$, while $\det e$ changes sign. Hence, with f built from TetB, $g^{-1} f = I$, and the HR potential takes the same value on either root. Only the vielbein form $e^{-1}(Pe) = P$ sees the second root, and there $V(P) = 0$.*

Proof. $P^{\mathsf{T}} \eta P = \eta$; checked for a general static tetrad. □

The two roots are the two square-root branches of $g^{-1} f = I$: P satisfies $P^2 = I$ but is not the principal root (its spatial eigenvalues are -1), and $e^{\mathsf{T}} \eta (Pe)$ is symmetric, so the vielbein form with $S = P$ meets the symmetric-vielbein condition (checked). The metric form, defined with the principal root, sees only $S = I$.

This is Letter 2's Lemma [4]: objects built from the metric (the metric, a torsion-free connection) cannot tell TetA from TetB, and only the torsion sector can. The HR interaction is built from the metrics and contains no torsion, so it cannot carry the TetA/TetB partition. That is the reason for Theorems 1–4.

Part B: a deformed frame (diagnostic)

Theorem 6 (exchange closure on a deformed frame). *On $S = \mathrm{diag}(1, 1, 1, w)$, for every β,*

$$\begin{aligned} V &= A + Bw, \\ A &= \beta_0 + 3\beta_1 + 3\beta_2 + \beta_3, \\ B &= \beta_1 + 3\beta_2 + 3\beta_3 + \beta_4. \end{aligned}$$

The deformed leg's share $s(w) = Bw/(A + Bw)$ satisfies $s(1/w) = 1 - s(w)$ if and only if $A = B$, that is

$$\beta_0 + 2\beta_1 = \beta_4 + 2\beta_3, \tag{2}$$

and then $s(w) = w/(1 + w)$ exactly.

Proof. $s(1/w) - [1 - s(w)] = -w(A - B)(A + B)/[(A + Bw)(Aw + B)]$ (checked), which vanishes iff $A = B$ since $A + B > 0$. □

The share is not an arbitrary choice: $Bw/(A + Bw) = w\,\mathrm{d}\ln V/\mathrm{d}w$, the elasticity of V on this family (checked).

Corollary 1 (the lock and β_2). *HR's $g \leftrightarrow f$ symmetry $\det S\, V(S^{-1}) = V(S)$ holds iff β is palindromic (the difference is $\sum_n (\beta_{4-n} - \beta_n) e_n(S)$). Palindromic β satisfies (2), and then $V = (\beta_0 + 4\beta_1 + 3\beta_2)(1 + w)$. The locked set gives $V = 49(1 + w)$, and β_2 cancels from s. On this family $V/V(I) = \det[(I + S)/2]$, the volume of the midpoint frame.*

Proof. All identities checked exactly; $(1, 0, 16, 0, 2)$ shows that (2) is a real condition. □

The share's form follows from the $g \leftrightarrow f$ symmetry, not from the value 16.

Remark (not the two chambers). The two shares $w/(1+w)$ and $1/(1+w)$ sum to 1, have odds w, and swap under $w \to 1/w$: the algebra of the partition. But they are the deformed leg and the three undeformed legs of one frame, not TetA and TetB. On the actual roots the locked potential is $V(I) = 98$ and $V(P) = 0$, and on the path between them, $\mathrm{diag}(1, s, s, s)$, it is $(1+s)(s^2+47s+1)$, with no exchange symmetry between the ends (checked). So this is not the two-chamber mechanism of [9, 4].

Theorem 7 (the time leg, and what the potential does not supply). *(i) With $w = |a|/a_0$, a the acceleration of the tetrad's time leg, a static field has $c^2|a| = |\nabla\Phi|$ exactly [8], so $w = x$ and $s = \mu(x)$. (ii) The auxiliary energy*

$$E(u; y) = (1 - e^{-u})\, y - 2(\sinh u - u),$$

with $u = \ln(V/V_0)$, is stationary at $V/V_0 = 1 + \sqrt{y}$, where its value is $F(y)$ and its slope $\partial E/\partial y$ is μ. (iii) Its u-dependence contains no β.

Proof. All checked exactly. □

So the potential supplies the variable $V/V_0 = 1 + x$; the energy F, and with it the dynamics, comes from the SST's own scalar term.

Conclusion and scope

- The locked HR theory does not reduce to $\mu(x)$ by the standard reduction (Theorems 1–3), and no choice of β would change that. Papers VIII, IX and XII overstate the equivalence.
- What the lock does fix is bimetric physics: the de Sitter background $\Lambda = m^2(1 + 48c^2)$, the Fierz–Pauli mass, the Higuchi bound and the screening cubic (1) [10].
- HR is root-blind (Theorem 5), so its two metrics are not GFD's two chambers.
- On a frame deformed along the time leg's acceleration the potential reproduces the partition's algebra (Theorem 6, Corollary 1, Theorem 7), but not its mechanism: the shares are not the two roots, the frame is not an HR reference metric, and F is not in V. The constitutive law's origin remains the two-root partition under exchange [4, 9].
- Letter 2 locates the partition in the torsion sector. A dual-chamber bridge would therefore couple connections, not metrics; that is the subject of [5].
- Not done here: the full bimetric (dynamical f) decoupling limit, which keeps the linear enhancement below 4/3; and a derivation of the deformation $w = |a|/a_0$ from an action.

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